

Joshua N. **Grant**
Geospatial Systems Architect

👤 Contact

📍 12918 Sanderling Lane
Knoxville, TN 37922
United States of America
📞 +1 (865) 803 3495

✉️ jngrant@live.com
B notjustadatum.blogspot.com
🔗 github.com/sempervent
in linkedin.com/in/joshuanagrant

🎯 Objective

To take theories, ideas, research, and experiments and turn them into production-scalable solutions while developing and supporting applications and data workflows on cloud infrastructure.

⚙️ Skills

🔗 **Programming Languages:** Python, R, JavaScript, SQL, Bash, PHP, Rust, $\LaTeX$

☁️ **Cloud Platforms:** AWS (S3, Lambda, EC2, RDS, Fargate), GCP, Azure

🗄️ **Databases:** PostgreSQL, MongoDB, MySQL, Spatialite, Snowflake, SQLite, DuckDB, Redis, RabbitMQ, MQTT

⚙️ **Data Engineering:** Airflow, Prefect, Kafka, Scala, Hadoop, TimescaleDB, NiFi, ETL Pipelines

📍 **Geospatial Tools:** ArcGIS, GeoPandas, Geoparquet, QGIS, GDAL, PostGIS, Martin, Pyproj, Rasterio, MapBox, Leaflet

▶️ **DevOps:** Docker, Kubernetes, Ansible, GitHub Actions, AWS CodePipeline

📊 **Data Visualization:** Shiny, Leaflet, Kibana, matplotlib, Plotly, Grafana

🏗️ **Frameworks and APIs:** FastAPI, Flask, Django, Swagger, REST API Development, Electron

⚡ **Machine Learning:** TensorFlow, Scikit-learn, Data Preprocessing, Model Deployment

🔧 **Version Control:** Git (GitHub, GitLab, Bitbucket)

👥 **Soft Skills:** Team Leadership, Agile Methodologies, Cross-Functional Collaboration, Written & Verbal Communication, Problem Solving

🏠 Projects

Final Fantasy Football — *A semantic learning algorithm for fantasy football*

Tech: Python, Scikit-learn, Pandas, NumPy, Matplotlib, Requests, BeautifulSoup, Flask, Docker, Docker Compose, Airflow

Where I've Been — *A web application to visualize travel history*

Tech: Python, Flask, Docker, Docker Compose, PostgreSQL, Leaflet, Vue, Bootstrap

This Is A Casino — *A web application to visualize and trade stocks via semantic learning, reinforcement learning, and deep learning*

Tech: Python, Flask, Docker, Docker Compose, PostgreSQL, Leaflet, React, Bootstrap

Genesis — *A python audio input game to create a universe from a sound*

Tech: Python, pygame

Decentralized Content Reward System — *A decentralized content reward system for the web*

Tech: Python, Rust, Flask, Docker, Docker Compose, PostgreSQL, Leaflet, React, Bootstrap

Cosmic Architect — *A pygame game where you compete to build the best planet*

Tech: Python, pygame

maw — *A cli for quickly combining tabular data into parquet*

Tech: Rust

Experience

Geospatial Systems Architect

Oak Ridge National Laboratory, Oak Ridge, TN 2023—*Present*

- Architecting and maintaining a geoparquet-based data warehouse for storage and retrieval of risk components
- Designing, architecting, and maintaining simultaneous deployment of a Geospatial Decision Support Tool for both web and desktop environments
- Architecting a geospatial data pipeline, using Airflow for the ingestion, storage, and analysis of raster data in a cloud environment
- Developing geospatial applications for the tracking and monitoring of high-sensitivity parcels using IoT technologies, Kafka, TimescaleDB, and Airflow running in AWS
- Designing and architecting a geospatial data pipeline, database, frontend, and backend for the ingestion, storage, and retrieval of clock drift timeseries data using Airflow
- Developing and maintaining a geospatial data warehouse and pipeline for the ingestion, storage, and retrieval of raster data with data ingestion controlled by Prefect
- Architecting a geospatial database capable of handling large-scale raster data and aggregating zonal stats to a variety of vector geometries

Senior Software Engineer Data Engineering

Bold Penguin, Columbus, OH (remote) 2022—2023

- Designed a more centralized system for development and deployment using GitHub Actions, AWS, and Python modules
- Supervised a team effort to migrate deployments, flows, and tasks from Prefect 1 to Prefect 2
- Developed Prefect tasks to run on either EC2 or Fargate instances based on conditions such as file size
- Created EC2 and Fargate microservices for assistance in document conversion and data extraction
- Designed and maintained REST APIs to be up-to-date with stakeholder and client demands
- Designed and implemented a scalable message queue (MQ) system to handle high load for a data extraction service, resulting in improved performance and increased efficiency of data processing operations
- Configured GitHub Actions and AWS CodePipeline to automate the build, test, and deployment process of the projects, ensuring consistent and reliable delivery of software updates
- Created and oversaw the deployment of multiple Python and JavaScript projects using AWS CodePipeline, resulting in streamlined, automated, and efficient project delivery

Data Engineer

Oak Ridge National Laboratory, Oak Ridge, TN 2019—2022

- Served as support staff for a Python-based web scraping project, implementing an ETL workflow from a brittle-based cron system to a more robust Airflow system deployed in a Kubernetes cluster and in the cloud
- Developed and maintained full stack docker images using Python, R, Airflow, and Shiny to automate the acquisition of spatial-temporal data for the WSTAMP project for storage in an Hadoop Data Warehouse
- Led the development team for a COVID-19 tracking project using both R and Python and downstream model creation and analysis
- Developed containerized Airflow ETL workflows, Python microservices, and REST APIs using Swagger for data exchange with PostgreSQL, AWS S3 & RDS, and MongoDB
- Automated web-scraping and social media ingestion via containerized model creation for misinformation classification for implementation in advanced statistical models and network analysis and ingestion into ElasticSearch and visualization in Kibana
- Served as Software Quality Assurance board chair for the Geospatial Science and Human Security Division
- Participated as a member of the ORNL Data Council, which discusses best practices in lab-wide data management and architecture for sharing of lab-generated data

Data Scientist II Subcontractor

Oak Ridge National Laboratory, Oak Ridge, TN 2018—2019

- Automated data ingestion of the WSTAMP Web Application via the use of R, Python, Airflow, and Docker
- Authored an R package to unify geography between online data APIs
- Developed a SpatiaLite SQLite database to ease lookup of geographic data
- Developed a zonal statistics application in Python that takes a raster image and shapefile and computes the statistics over the shapes stored in the shapefile

Owner & Contractor-For-Hire

Specrabella, Knoxville, TN 2018—2018

- Designed, developed, and deployed an interactive energy informatics leaflet-based web application integrated with Snowflake that provides real-time calculation from machine learning derived models
- Consulted with clients on ground-up web application architecture, database and visualization needs
- Developed data ingestion methods using cURL, PHP, and R to scrape energy data from the web
- Streamlined training and development procedures using ansible, vagrant, PHP, and BASH
- Designed and implemented a data pipeline using R, Python, and Snowflake to ingest, clean, and visualize energy data
- Developed a web application to visualize energy data using R, Shiny, and Leaflet
- Architected, designed, and maintained PostgreSQL databases for energy and financial data

Bioinformatician, Project Manager, & Technical Sales Representative

Microbial Insights, Knoxville, TN 2016—2018

- Established an automated pipeline using R, MySQL, Python, and Bash for NGS analysis, along with intranet website control written in JavaScript and Shiny
- Automated client-bound statistical calculations such as linear models, ANOVA, SOMs, clustering, PCA, and PCoA
- Developed an ETL customer data visualization tool using R, PHP, JQuery, and MySQL to view qPCR results in the context of other samples' and dynamically selectable parameters
- Optimized data delivery to clients via a custom R package and local shiny applications to quickly generate $\LaTeX$ PDF reports
- Constructed, populated, and maintained an intranet wiki using PostgreSQL and PHP to aid in project management and customer service

Graduate Research Assistant, Dr. Neal Stewart's Plant Biotechnology Lab

University of Tennessee, Knoxville, TN 2014—2016

- Summarized statistical findings of cell suspensions using linear models, ANOVA, and PCA in Python and R
- Evaluated suspension cultures via chemical and spectral processes for lignin formation and statistically analyzed and summarized my findings for inclusion in DOE reports
- Collaborated on a novel single cell suspension and cryopreservation robotic system
- Advanced and executed monocot genetic modifications for *Panicum* spp., *Oryza* spp., *Sorghum* spp., *Saccharum* spp., *Zea mays*

Laboratory Assistant, Dr. Neal Stewart's Plant Biotechnology Lab

University of Tennessee, Knoxville, TN 2012—2014

- Developed automated statistical methodology for screening of lignin content and imaging of cell characteristics both *in vivo* and *in vitro* using Python
- Extracted genomes from NCBI, cleaned and normalized the data, and performed exploratory data analysis using Python
- Complied with USDA-APHIS regulations regarding transgenic plants
- Implemented an *E. coli* bioreactor for production of proteins

Laboratory Assistant, Dr. Paris Lambdin's Biosystematics and Biological Control Lab

University of Tennessee, Knoxville, TN 2002—2012

- Designed instructional modules for undergraduate and graduate level courses using HTML, CSS, JavaScript, and Flash
- Collected samples and data from field locations
- Assisted in community outreach programs such as Bloomsdays, Buggy Buffet, and 4-H Camps
- Performed forest coverage analysis using ArcGIS

Education

Master of Science in Plant Sciences —Plant Molecular Genetics

University of Tennessee

Spring 2017, GPA: 3.72/4.0

Bachelor of Science in Plant Sciences —Biotechnology

University of Tennessee

Spring 2014, GPA: 3.74/4.0 —Magna Cum Laude

Publications

- [1] N. Labbe L.M. Kline M. T. Windham D.G.J Mann C. Fu J.N. Grant C.R. Poovaiah H. Shen W. A. Wuddineh A. Ziebell M. F. Davis F. Chen T. J. Tschaplinski J. R. Mielenz Z-Y Wang R. A. Dixon C. N. Stewart Jr Baxter, H.L. M. Mazarei. Field-based experiments on low-lignin switchgrass as a feedstock for lignocellulosic biofuel production. 2012.
- [2] N. Labbe L.M. Kline M. T. Windham D.G.J Mann C. Fu J.N. Grant C.R. Poovaiah H. Shen W. A. Wuddineh A. Ziebell M. F. Davis F. Chen T. J. Tschaplinski J. R. Mielenz Z-Y Wang R. A. Dixon C. N. Stewart Jr Baxter, H.L. M. Mazarei. Field-based experiments on low-lignin switchgrass as a feedstock for lignocellulosic biofuel production. 2012.
- [3] Christa Brelsford, Sarah Tennille, Aaron Myers, Supriya Chinthavali, Varisara Tansakul, Matthew Denman, Mark Coletti, Joshua Grant, Sangkeun Lee, Karl Allen, et al. A dataset of recorded electricity outages by united states county 2014–2022. *Scientific Data*, 11(1):271, 2024.
- [4] Joshua N Grant, Jason N Burris, C Neal Stewart, and Scott C Lenaghan. Improved tissue culture conditions for the emerging c 4 model *Panicum hallii*. *BMC biotechnology*, 17(1):39, 2017.
- [5] Joshua N. Grant, Jonathan D. Willis, and C. Neal Stewart. Functional analysis of a putative membrane-bound switchgrass endo- β -1,4-glucanase in *Escherichia coli*. 2012.
- [6] Joshua N Grant, Jonathan D Willis, and C Neal Stewart. Comparisons of switchgrass cell wall components containing an overexpression of a putative switchgrass endoglucanase. In *IN VITRO CELLULAR & DEVELOPMENTAL BIOLOGY-PLANT*, volume 49, pages 475–475. SPRINGER 233 SPRING ST, NEW YORK, NY 10013 USA, 2013.
- [7] Joshua Nathaniel Grant. Evaluation of hall’s panicgrass (*Panicum hallii* vasey) as a model system for genetic modification of recalcitrance in switchgrass (*Panicum virgatum* (L.)). Master’s thesis, 2017.
- [8] Dasha Herrmannova, Gautam Thakur, Josh Grant, Varisara Tansakul, Bryan Eaton, Olivera Kotevska, Jordan Burdette, Martin Smyth, and Monica Smith. Challenges in automated detection of covid-19 misinformation.
- [9] Jinyi Liu, Nana Chen, Joshua N Grant, Zong-Ming Cheng, C Neal Stewart Jr, and Tarek Hewezi. Soybean kinome: functional classification and gene expression patterns. *Journal of experimental botany*, 66(7):1919–1934, 2015.
- [10] Alina Peluso, Deeksha Rastogi, Hilda B. Klasky, Jeremy Logan, Dakotah Maguire, Josh Grant, Blair Christian, and Heidi A. Hanson. Environmental determinants of health: Measuring multiple physical environmental exposures at the united states census tract level. *Health & Place*, 89:103303, 2024.
- [11] Robert Stewart, Samantha Erwin, Jesse Piburn, Nicholas Nagle, Jason Kaufman, Alina Peluso, J. Blair Christian, Joshua Grant, Alexandre Sorokine, and Budhendra Bhaduri. Near real time monitoring and forecasting for covid-19 situational awareness. *Applied Geography*, 146:102759, 2022.
- [12] Robert Stewart, Jesse Piburn, Sarah Walters, Jason Kaufman, Evan Ezell, David Anderson, David Axley, Josh Grant, Bryan Eaton, Alexandre Sorokine, et al. A taxonomic classification approach for global spatio-temporal data. 2021.
- [13] Varisara Tansakul, Aaron Myers, Sarah Tennille, Matthew Denman, Alec Hamaker, Jonathan Huihui, Kyle Medlen, Karl Allen, Daniel Redmon, Supriya Chinthavali, et al. Eagle-i power outage data 2014-2022. Technical report, Oak Ridge National Lab.(ORNL), Oak Ridge, TN (United States). Oak Ridge . . . , 2023.
- [14] Varisara Tansakul, Aaron Myers, Sarah Tennille, Matthew Denman, Alec Hamaker, Jonathan Huihui, Kyle Medlen, Karl Allen, Daniel Redmon, Supriya Chinthavali, et al. Eagle-i power outage data information. Technical report, Oak Ridge National Lab.(ORNL), Oak Ridge, TN (United States). Oak Ridge . . . , 2023.
- [15] Jonathan D Willis, Joshua N Grant, Mitra Mazarei, Lindsey M Kline, Caroline S Rempe, A Grace Collins, Geoffrey B Turner, Stephen R Decker, Robert W Sykes, Mark F Davis, et al. The tceg1 beetle (*tribolium castaneum*) cellulase produced in transgenic switchgrass is active at alkaline ph and auto-hydrolyzes biomass for increased cellobiose release. *Biotechnology for Biofuels*, 10(1):230, 2017.

References

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